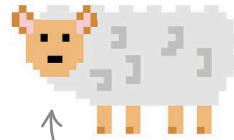




LINGO

Integers and floats

In coding, whole numbers are called integers, while numbers with a decimal point in them are known as floats. Programs usually count things using integers. Floats are more often used for measurements.



1 sheep (an integer)



0.5 sheep
(a float)

Using numbers

Variables can be used to store numbers and do sums. You can use them with symbols to do calculations, just like you do in maths. Some of these symbols will be familiar, but watch out for the symbols meaning “multiply” and “divide” – they’re slightly different from the ones you use in class.

Symbol	Meaning
+	add
–	subtract
*	multiply
/	divide

Some of the Python maths symbols

1 A simple calculation

Type this code in a shell window. It uses numbers stored in two variables, named **x** and **y**, to carry out a simple multiplication. Hit the enter/return key to get the answer.

```
>>> x = 6
>>> y = x * 7
>>> print(y)
42
```

Create a new variable, **x**, and give it the value 6.

Print the value of **y**.

Multiply **x** by 7 and store the result in **y**.

The result of the calculation

Change the value of **x**.

2 Change a value

To change the value of a variable, you just assign a new value to it. In your code, change the value of **x** to 10 and run the calculation again. What do you expect the result to be?

```
>>> x = 10
>>> print(y)
42
```

The result hasn't changed – next we'll find out why.

Update the value of **y**.

3 Update the value

The value of **y** needs to be updated to get the correct result. Type these lines. Now the code assigns the new value to **y** after **x** has been changed. If you update the value of one variable in your own programs, always check to see if you need to update any others.

```
>>> x = 10
>>> y = x * 7
>>> print(y)
70
```